Neeraj Kumar

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EDUCATION

Guru Gobind Singh Indraprastha University - USAR **Nov 2022 – Jul 2026**
Bachelor's Degree - Automation & Robotics - **CGPA - 7.00** New Delhi

TECHNICAL SKILLS

Languages: C++17/20, C, Python, JavaScript
Technologies/Frameworks: React, Node.js, Express, PostgreSQL, MongoDB, Redis, WebSockets, OpenSSH, Docker, Linux/UNIX environments, REST APIs, TCP/IP
Tools: Git, Nginx, GitHub Actions, Vercel, Postman, VS Code
Core Competencies: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, System Design, Distributed Systems

EXPERIENCE

Outlier **Mar 2026 – Apr 2026**
AI Code Reviewer & Data Annotator Remote

- Reviewed and evaluated AI-generated code and technical responses for correctness, logic, edge cases, and adherence to task requirements.
- Annotated and ranked model outputs, identifying logical errors, inefficient implementations, and inconsistencies to improve code-generation quality.
- Applied structured evaluation criteria across programming and technical tasks, providing detailed feedback and corrected solutions.

Open Source Contributions
Contributor Remote

- Contributed code and documentation fixes to **aardappel/Lobster** and **Garudex-Labs/Caracal**, improving code quality and resolving inconsistencies between documentation and implementation.
- Contributed to **shogunlabs/augur**, improving IPO data pipeline reliability through retry logic, asynchronous data hydration, health-check endpoints, and PostgreSQL export workflows.
- Merged **10+ pull requests** across open-source repositories, including **knadh/pkg.bot**; fixed repository search tests and Repology database download handling, and continue contributing bug fixes and improvements to community-maintained projects.

PROJECTS

FixEngine | **C++20, CMake, Linux, TCP/IP** | **github.com/drunktrader/FixEngine** **May 2026 – Aug 2026**

- Built a low-allocation FIX protocol parsing engine in **C++20** with a 6-stage pipeline spanning ingestion, tokenization, parsing, validation, formatting, and orchestration.
- Designed bounded, zero-copy message storage using fixed-capacity containers and `std::string_view`; benchmarked core operations at approximately ~ 2 **ns/op**.
- Implemented **checksum** and **body length validation** with explicit error types, maintaining deterministic, exception-free control flow on the normal processing path.

RedisCLI | **C++17, POSIX Sockets, RESP2** | **github.com/drunktrader/RedisCLI** **Mar 2026 – Apr 2026**

- Built a Redis CLI client from scratch in **C++17**, implementing the **RESP2** protocol **serializer** and **parser**.
- Implemented event-driven **Pub/Sub** mode using **poll()** to multiplex stdin and Redis socket simultaneously - streaming published messages while keeping terminal input responsive via GNU Readline callback API.

Options Pricing Simulator | **Python** | **github.com/drunktrader/options-pricing-model** **Feb 2026 – Mar 2026**

- Developed a **Monte Carlo** simulation engine to price **European Options** with configurable simulation runs.
- Validated results against Black-Scholes analytical model, achieving $<$**0.03** pricing deviation at **10,000** simulations.
- Computed option sensitivities : Delta (Δ), Vega (ν), Gamma (Γ), and Theta (Θ) - to support quantitative risk analysis.

Discord-Vault | **Python, SHA256** | **github.com/drunktrader/discord-vault** **Jul 2026 – Sep 2026**

- Developed a secure Discord-based file vault for storing and retrieving encrypted files through a bot interface.
- Implemented **SHA-256** based integrity verification to detect file tampering and ensure data consistency.
- Implemented lossless PNG compression and decompression, reducing image storage overhead while preserving pixel-perfect data integrity.
- Automated file upload, retrieval, and vault management through Discord commands, eliminating the need for a dedicated frontend.

CERTIFICATIONS

- **Postman API Fundamentals Student Expert** - Postman
- **Career Essentials in GitHub Professional Certificate** - LinkedIn Learning
- **Agentic AI Certified Foundations Associate** - Oracle University

ACHIEVEMENTS & EXTRACURRICULARS

- Achieved Max Rating of **1614** on **LeetCode**; solved **300+** algorithmic problems on LeetCode in C++.
- Achieved Global Rating of **123** on CodeChefs and max rating of **1159** on **Codeforces**.
- Represented the college Volleyball team in inter-collegiate tournaments and secured **3rd place**.